Elementary Complex Analysis

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Abstract

This is a text of elementary complex analysis referencing "COMPLEX ANALYSIS" of Elias Stein and Rami Shakarchi.



Figure 1: Elias Stein

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Chapter 1

Complex Numbers and Complex Functions

1.1 Complex Numbers

Before systematically learning complex analysis, we have to first understand the complex numbers $\mathbb{C}$. One might have heard that the complex numbers were introduced to solve the problems of the non-real roots of polynomial equations. For instance that considering the equation

$$0 = x^2 + 1$$

In reals, it is impossible for certain $x_0 \in \mathbb{R}$ such that its square is negative. Thus, we introduce the imaginary unit i defined by

$$i^2 = -1$$

and we endow the scalar multiplication for yi where $y \in \mathbb{R}$. Then, we define the complex numbers by the forms of $z = x + yi$ where $x, y \in \mathbb{R}$ and called the real Part and the imaginary Part of z respectively, denoted by $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$.

The operations of complex numbers are defined by the respective operations of the real Part and the imaginary Part. If $z = x + yi$ and $w = a + bi$, then

$$z + w = (\mathrm{Re}(z) + \mathrm{Re}(w)) + (\mathrm{Im}(z) + \mathrm{Im}(w))i = (x + a) + (y + b)i$$

$$z\ w = (x + yi)(a + bi) = (xa - yb) + (xb + ya)i$$

$$\frac{z}{w} = \frac{x + yi}{a + bi} = \left(\frac{x + yi}{a + bi}\right)\left(\frac{a - bi}{a - bi}\right) = \frac{1}{a^2 + b^2}\left((xa + yb) + (ya - xb)i\right)$$

Based on the operations above, we can easily deduce the algebraic properties of $\mathbb{C}$, for $z_1, z_2, z_3 \in \mathbb{C}$:

- Commutativity: $z_1 + z_2 = z_2 + z_1, z_1 z_2 = z_2 z_1$
- Associativity: $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3), (z_1 z_2) z_3 = z_1 (z_2 z_3)$
- Distributivity: $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$

Note that we can use an ordered pair $(x, y) \in \mathbb{R}^2$ to uniquely determinea a complex number $x + yi$ so that in terms of topology, $\mathbb{C}$ is homeomorphic to $\mathbb{R}^2$ and we can use a plane to represent $\mathbb{C}$, called complex plane. Therefore, we can directly apply the topology of $\mathbb{R}^2$ to $\mathbb{C}$, such as convergence, connectedness, compactness, and etc.

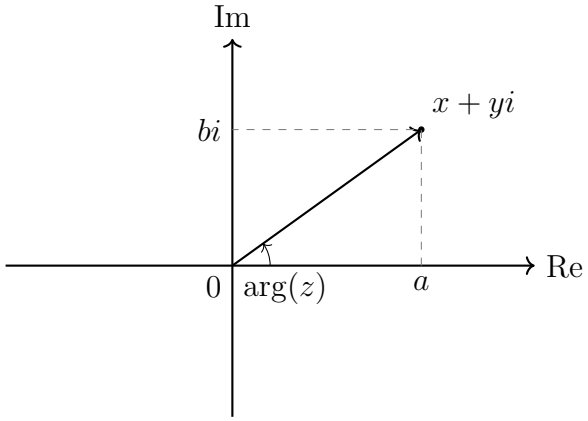


Figure 1.1: Complex Plane

Next, we introduce some important terms of complex numbers. The magnitude of a complex number $z = x + yi$, denoted by $|z|$, is defined by

$$|z| = \sqrt{x^2 + y^2}$$

The argument of z , denoted by $\arg(z)$ is defined by the angle between z and the real axis

$$\arg(z) = \arctan \frac{y}{x}$$

The conjugate of z , denoted by $\bar{z}$, is a complex number such that

$$\bar{z} = x - yi$$

which is exactly the complex number with the opposite imaginary Part of z . Clearly, if $z \in \mathbb{R}$, $\bar{z} = z$ and if $z = yi$, $\bar{z} = -z$. And, one should notice that $|z|^2 = z\bar{z}$.

A useful and prominent result found by Euler is the polar form of the complex numbers, that since

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}, \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

and

$$z = x + yi = |z| \left(\frac{x}{|z|} + \frac{y}{|z|} i \right) = |z|(\cos\theta + i\sin\theta)$$

we have

$$z = |z| e^{i\theta}$$

where $\theta = \arg(z)$. If z is unit, then

$$e^{ix} = \cos x + i \sin x$$

which is called the Euler Formula, playing a pivotal role in Math.

Let $z = re^{i\varphi}$ and $w = se^{i\theta}$, then

$$zw = rs\ e^{i(\varphi+\theta)}$$

so that the multiplication of the complex numbers actualy conducts a rotation in the plane.

1.2 Complex Functions

It is no surprise for those who have learned the multivariable functions when first encountering the complex functions. The formal form ofthe complex functions are the maps $f : \mathbb{C} \longrightarrow \mathbb{C}, z \mapsto f(z)$. Since the complex numbers have actually two variables, which are the real and the imaginary Part, the complex functions are essentially a type of two-variable functions. However, the definition that $i^2 = -1$ make them be distinct from the general two-variable functions.

We first introduce the continuous functions, which is defined as usual.

Definition 1.2.1: Continuous functions
Let f be a function defined on a set $\Omega \subset \mathbb{C}$. Then, f is continuous at $z_0 \in \Omega$ if $\forall \varepsilon > 0, \exists \delta > 0$ such that $\forall z \in \Omega$,

$$|z - z_0| < \delta \implies |f(z) - f(z_0)| < \varepsilon$$

f is continuous on Ω if f is continuous at all $z \in \Omega$.

Since $\mathbb{C} \cong \mathbb{R}^2$, we can also express the definition above by the form of balls.

$$\forall \varepsilon > 0, \exists \delta > 0, \forall z \in \Omega, z \in B_\delta(z_0) \implies f(z) \in B_\varepsilon(f(z_0))$$

Moreover, an equivalent definition of continuity is by the limit of sequence that if f is continuous at z_0 , then

$$\forall \{z_n\}_{n=1}^{\infty} \subset \Omega, \lim_{n\to\infty} z_n = z_0 \implies \lim_{n\to\infty} f(z_n) = f(z_0)$$

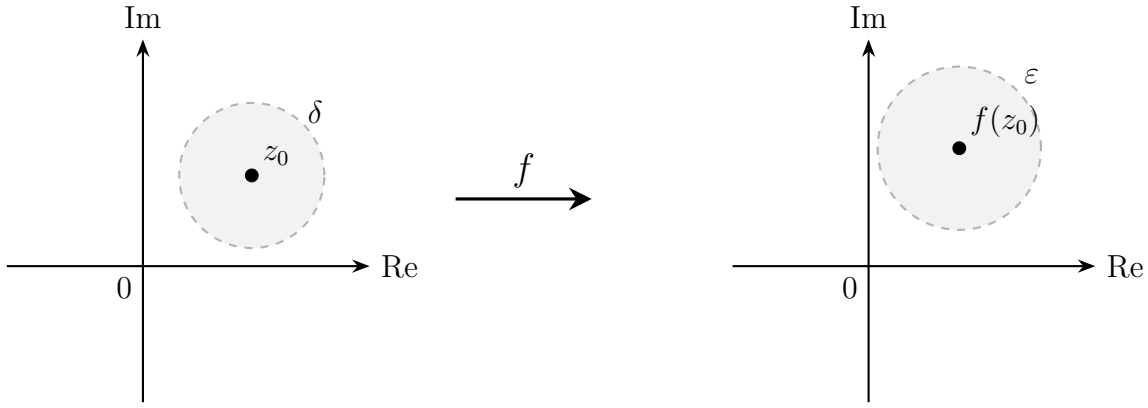


Figure 1.2: Visualisation of continuous complex functions

For real continuous functions, we have known that they always attain the extreme in a compact set. We can define the similar concept by the magnitude of the functions that f attains a maximum at $z_0 \in \Omega$ if

$$\forall z \in \Omega, |f(z)| \leqslant |f(z_0)|$$

with the inequality reversed for the definition of a minimum.

Note that by the triangle inequality, $|f|$ is continuous if f is continuous. Indeed

$$| \, |f(z_0)| - |f(z)| \, | \leqslant |f(z_0) - f(z)|$$

We can deduce a plenty of properties of the magnitude of the complex functions according to the continuity of $|f(z)|$.

Theorem 1.2.1:
A continuous functions on a compact set Ω is bounded and attains a maximum and minimum on Ω .

Theorem 1.2.2:
Let f be a continuous function on $\Omega \subset \mathbb{C}$, and denote by $M = \max_{z\in\Omega} |f(z)|$, $m = \min_{z\in\Omega} |f(z)|$. Then,

$$\forall y \in [m, M], \exists z_0 \in \Omega : |f(z_0)| = y$$

The theorems here follow immediately from the properties of single-variable continuous functions.

1.3 Holomorphic Functions

Now, we discuss the derivative of complex functions.

Definition 1.3.1: Holomorphic Functions
Let Ω be an open set in $\mathbb{C}$ and f a complex-valued function on Ω . Then, f is holomorphic at $z_0 \in \Omega$ if the limit

$$\lim_{h\to 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

converges where $h \in \mathbb{C}$ and $z_0 + h \in \Omega$, and the limit is called the derivative of $f(z)$ at z_0 , denoted by $f'(z_0)$. f is holomorphic on Ω if f is holomorphic at every point of Ω .

The terms "complex differentiable" or "regular" are same to holomorphic.

Definition 1.3.2: Entire Functions
If $f(z)$ is holomorphic at all $z \in \mathbb{C}$, then $f(z)$ is an entire function.

In the following sections, you will see the advantages of holomorphic functions than differentiable function of single real variable. For example, the holomorphic functions are actually infinitely many times complex differentiable; the holomorphic functions can always be expanded to a power series so that we usually use the term "analytic" as a synonym for holomorphic.

A basic property of the holomorphic functions is its closure under the elementary operations.

Proposition 1.3.1:
If f and g are holomorphic in Ω , then

- $f + g$ is holomorphic in Ω and $(f + g)' = f' + g'$.

- $f\ g$ is holomorphic in Ω and $f\ g = f'g + g'f$.
- If $g(z_0) \neq 0$, then $\frac{f}{g}$ is holomorphic at z_0 and

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

- If $f : \Omega \longrightarrow U$ and $g : U \longrightarrow \mathbb{C}$ are holomorphic, then the chain rule holds and

$$\forall z \in \Omega,\ (g \circ f)' = g'(f(z))f'(z)$$

As the complex numbers are consisted of the ordered pair $(x, y) \in \mathbb{R}^2$, we can also write the complex functions with following form by $z = x + yi$

$$f(x, y) = u(x, y) + v(x, y)i$$

where $u(x, y) = \mathrm{Re}(f(z)), v(x, y) = \mathrm{Im}(f(z))$. Then, we can take the partial derivatives of f as a multi-variable function

$$\frac{\partial f}{\partial x} = \frac{\partial u(x, y)}{\partial x} + i\frac{\partial v(x, y)}{\partial x}$$

$$\frac{\partial f}{\partial y} = \frac{\partial u(x, y)}{\partial y} + i\frac{\partial v(x, y)}{\partial y}$$